Introduction to LaTeX (Mathematical notations,Equations,Array,Matrices,List)

Dr. Neeta Nain

MNIT Jaipur

Inline vs Dispaly Math

Math in LaTeX starts and ends with $...$ or \(...\) for inline, \[...\] for display

Editor View:

LaTeX Code

```
Inline: \( x+y=10 \)
Display:
\[
x+y=10
\]
```

Document View:

Inline: $x + y = 10$

Display :

$$x + y = 10$$

Explanation:

- Inline math uses `\( ... \)`.
- Display math uses `\[ ... \]` for centered equations.

– Superscripts and Subscripts

Superscripts use ^ and subscripts use

Editor View:

LaTeX Code

```
$a_i^2 + b_j^3$
```

Document View: $a_i^2 + b_j^3$

–Greek Letters and Special Symbols

Greek letters use backslash then letter name

Editor View:

LaTeX Code

```
$\alpha, \beta, \gamma, \delta, \lambda, \pi$
```

Document View: $\alpha, \beta, \gamma, \delta, \lambda, \pi$

– Fractions and Roots

Editor View:

LaTeX Code

```
$\frac{a+b}{c}$, $\sqrt{x}$, $\sqrt[3]{y}$
```

Document View: $\frac{a+b}{c}$, $\sqrt{x}$, $\sqrt[3]{y}$

– Sums, Products, and Integrals

Editor View:

LaTeX Code

```
$\sum_{i=1}^{n} i$, $\prod_{k=1}^{m} k$,
$\int_0^\infty e^{-x} dx$
```

Document View: $\sum_{i=1}^{n} i$, $\prod_{k=1}^{m} k$, $\int_0^{\infty} e^{-x} dx$

– Relations and Logic Symbols

Math relation and logic

Editor View:

LaTeX Code

```
$a \leq b$, $x \neq y$, $A \cap B$, $A \cup B$, $A \
```

Document View: $a \leq b$, $x \neq y$, $A \cap B$, $A \cup B$, $A \subseteq B$

Numbered Equations

Editor View:

LaTeX Code

```
\begin{equation}
\int_0^1 x^2 \, dx = \frac{1}{3}
\end{equation}
```

Document View:

$$\int_0^1 x^2 \, dx = \frac{1}{3} \tag{1}$$

Explanation:

- `\frac{}{}` creates a fraction.
- `\int` adds an integral sign.
- Numbering is added using the `equation` environment.

Array

Using array environment to create aligned columns with custom alignment
Editor View:

LaTeX Code

```
\[
\begin{array}{lcr}
\text{Left} & \text{Center} & \text{Right} \\
a & b & c \\
1 & 2 & 3 \\
\end{array}
\]
```

Document View:

Left	Center	Right
a	b	c
1	2	3

Piecewise Functions with array

Editor View:

LaTeX Code

```
\[
f(x) = \left\{
\begin{array}{ll}
-x, & x < 0 \\
0, & x = 0 \\
x^2, & x > 0
\end{array}
\right.
\]
```

Document View:

$$f(x) = \begin{cases} -x, & x < 0 \\ 0, & x = 0 \\ x^2, & x > 0 \end{cases}$$

Explanation:

- Arrays are frequently used to represent piecewise (case-based) functions.
- Here, a left curly brace is sized automatically with left and right to bracket the cases.

Basic Matrix Without Delimiters

Editor View:

LaTeX Code

```
\[
\begin{matrix}
1 & 2 \\
3 & 4 \\
\end{matrix}
\]
```

Document View:

$$\begin{matrix} 1 & 2 \\ 3 & 4 \end{matrix}$$

Explanation: The basic matrix environment creates a matrix without any enclosing brackets. It aligns items in rows and columns with & and \\.

Matrices With Different Delimiters

Explanation:
The amsmath package offers several matrix environments differing by the delimiters used around the matrix:

- pmatrix (parentheses)
- bmatrix (square brackets)
- Bmatrix (curly braces)
- vmatrix (single vertical bars, for determinants)
- Vmatrix (double vertical bars, often for norms)

Editor View:

LaTeX Code

```
% Parentheses
\[
\begin{pmatrix}
1 & 0 \\
0 & 1 \\
\end{pmatrix}
\]

% Square brackets
\[
\begin{bmatrix}
a & b \\
c & d \\
\end{bmatrix}
\]
```

LaTeX Code

```
% Curly braces
\[
\begin{Bmatrix}
x & y \\
z & w \\
\end{Bmatrix}
\]

% Single vertical bars (determinant)
\[
\begin{vmatrix}
1 & 2 \\
3 & 4 \\
\end{vmatrix}
\]
```

LaTeX Code

```
% Double vertical bars (norm)
\[
\begin{Vmatrix}
5 & 6 \\
7 & 8 \\
\end{Vmatrix}
\]
```

Document View:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\begin{Bmatrix} x & y \\ z & w \end{Bmatrix}$$

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

$$\begin{Vmatrix} 5 & 6 \\ 7 & 8 \end{Vmatrix}$$

Larger Matrices and Extra Spacing

Editor View:

LaTeX Code

```
\[     A = \begin{bmatrix}
1 & 2 & 3 \\
4 & 5 & 6 \\
7 & 8 & 9 \\
\end{bmatrix}
\]
% Adding extra space between rows
\[     B = \begin{bmatrix}
1 & 2 & 3 \\[6pt]
4 & 5 & 6 \\[6pt]
7 & 8 & 9 \\
\end{bmatrix}
\]
```

Document View:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Explanation:

- can create larger matrices with as many rows and columns as needed.
- If you want to add vertical spacing between rows, add square brackets after the \\ to specify extra space (e.g., [6pt]).

Small Matrices for Inline Use

Editor View:

LaTeX Code

```
\[
\left( \begin{smallmatrix}
a & b \\
c & d \\
\end{smallmatrix} \right)
\]
```

Document View:

$$\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix} \right)$$

Explanation:

For inline math (inside text), use smallmatrix inside `\left( ... \right)` to create compact matrices that don't disrupt line spacing.

Lists

Explanation:

- **itemize:** Creates an unordered list with bullet points.
- **enumerate:** Creates an ordered (numbered) list.
- **description:** Creates a list for definitions or labeled items where each item has a term and a description.

Editor View:

LaTeX Code

```
% Unordered list with itemize
\begin{itemize}
    \item Apples
    \item Oranges
    \item Bananas
\end{itemize}
```

LaTeX Code

```
% Ordered list with enumerate
\begin{enumerate}
    \item First step
    \item Second step
    \item Third step
\end{enumerate}

% Description list
\begin{description}
    \item[HTML] HyperText Markup Language
    \item[CSS] Cascading Style Sheets
    \item[JS] JavaScript
\end{description}
```

Document View:

- Apples
- Oranges
- Bananas

1. First step
2. Second step
3. Third step

HTML HyperText Markup Language

CSS Cascading Style Sheets

JS JavaScript

Thank You!!!!!